
ControlledRAG: A Pre-Registered Audit and Minimum Reporting Standard for RAG Faithfulness Evaluation

Anonymous Author(s)
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Abstract

1 Retrieval-augmented generation (RAG) systems are typically evaluated using single
2 configurations of retriever, generator, metric, and threshold, yielding faithfulness
3 claims that are difficult to reproduce and compare across studies. We introduce
4 ControlledRAG, a pre-registered audit framework and minimum reporting standard
5 that exposes seven critical axes of variation in RAG evaluation: generator, retriever,
6 context structure, scorer, human calibration, threshold transfer, and cost. Using a
7 controlled pipeline with fixed generations and matched-context diagnostics, we
8 show that commonly reported improvements are often under-identified: the same
9 outputs can yield substantially different effect sizes across reasonable faithfulness
10 metrics, with low cross-scorer agreement and limited alignment with human
11 judgments. A compact second-generator replication preserves these discrepancies,
12 while threshold-transfer and cost-aware analyses reveal instability across datasets
13 and deployment settings. All experiments are pre-registered and released with
14 auditable per-query artifacts to enable exact reproduction. Rather than proposing
15 a new model or metric, ControlledRAG reframes RAG evaluation as a multi-axis
16 evidence problem and provides a standardized protocol for reporting faithful, repro-
17 ducible, and comparable results. Code: [https://anonymous.4open.science/](https://anonymous.4open.science/r/rag-hallucination-detection-MS/README.md)
18 [r/rag-hallucination-detection-MS/README.md](https://anonymous.4open.science/r/rag-hallucination-detection-MS/README.md)

19 1 Introduction

20 Retrieval-augmented generation (RAG) [Lewis et al., 2020] now powers deployed systems across
21 medical QA, legal research, and customer support, yet many papers still evaluate faithfulness with
22 a narrow template: one retriever, one generator, one scorer, one threshold, and one benchmark.
23 That template is convenient, but it *under-identifies* the claim. A result that looks decisive in one
24 generator/retriever/scorer cell can change sign, shrink, or become a deployment tradeoff under an
25 equally defensible audit choice.

26 The trigger for this paper is a previously published refinement-paradox claim of the form “improving
27 per-passage retrieval reduces answer faithfulness.”¹ The headline was interesting enough to audit, but
28 it does not survive the controls reported here. At scale, the original SQuAD generator-7B contrast
29 shrinks to 0.661/0.650/0.661 across no-refinement, aggressive-refinement, and gated-refinement
30 conditions; a matched-similarity intervention on context coherence is null; threshold-transfer cells
31 change sign; and a cost-aware head-to-head produces no dominating system.

32 The audit reframes the original headline as a methodological result: single-cell RAG faithfulness
33 numbers hide too many degrees of freedom. Each empirical result below corresponds to an axis that
34 a RAG faithfulness claim must disclose.

¹Cited at submission as a public claim; the preprint reference is omitted here for double-blind review and will be added on acceptance.

The positive contribution is **ControlledRAG**: a reusable audit checklist and minimum reporting standard for RAG faithfulness claims. ControlledRAG is not a new RAG method or leaderboard; it asks authors to vary or explicitly hold fixed seven axes—generator, retriever, context structure, scorer, human calibration, threshold transfer, and cost—so reviewers can see whether a claim is robust or axis-specific.

Contributions. We make four contributions, mapped to the seven-axis standard. (i) **The standard itself** (§4): a seven-axis disclosure checklist for RAG faithfulness claims, with one principle—each axis is a movable quantity that, if left unreported, lets a reader substitute a different reasonable value and reach a different conclusion. (ii) **A self-applied seven-axis audit** (§5–§10) on a previously published refinement-paradox claim, exercising every axis of the standard. (iii) **Two completed adjudicated human-eval slices** (§8): a typical-row calibration ($n=99$, $\kappa=0.774$) and a targeted disagreement slice ($n=100$, $\kappa=0.721$) that reach *different* scorer rankings; the slice-dependence is the load-bearing finding under the human-calibration axis. (iv) **An anonymized review artifact** containing auditable per-query CSVs, matched context pairs, both human-eval slices, source traces, and analysis scripts.

Why NeurIPS. ControlledRAG is methodological, not NLP-specific. The under-identification phenomenon is a measurement-theory issue for any LLM-judged evaluation: same fixed generations, different reasonable scorers, different reported contrasts. The seven-axis decomposition is the contribution. We *conjecture* the decomposition generalises to other LLM-as-judge settings (summarisation faithfulness, code-execution evaluation, chain-of-thought scoring); demonstrating that generalisation is out of scope here.

Roadmap. §2–§4 define the setting and standard; §5–§10 run the audit; §11–§13 summarize scope and implications. The supplementary material contains pre-specification, per-seed and per-cell details, context-conditioned NLI rescoring, both human-evaluation slices, the source trace, and related-work positioning.

2 Background and Related Work

A standard RAG system [Lewis et al., 2020, Gao et al., 2024] threads a dense retriever [Karpukhin et al., 2020, Wang et al., 2024], an optional reranker [Nogueira and Cho, 2019], an optional refinement step (e.g., Xu et al. [2024]), and a generator. The implicit assumption is that better local retrieval yields more faithful answers globally. Decoder-side correctors take a different position: Self-RAG [Asai et al., 2024] retrains the generator to emit reflection tokens, and CRAG [Yan et al., 2024] adds a retrieval evaluator that triggers web search; both pay latency cost. RAPTOR [Sarathi et al., 2024] clusters retrieved passages into a tree and retrieves from both leaves and LLM-summarised internal nodes (§10). Faithfulness is typically scored with NLI-style support metrics or LLM judges [Honovich et al., 2022, Laban et al., 2022, Es et al., 2024], occasionally with human annotation. Recent evaluation frameworks (ARES [Saad-Falcon et al., 2024], RAGChecker [Ru et al., 2024], RAGTruth [Niu et al., 2024], MIRAGE [Park et al., 2025], FaithJudge [Tamber et al., 2025]) supply judges, corpora, or benchmark suites; ControlledRAG is the disclosure layer around them. Discourse coherence has a long history in classical generation [Li and Hovy, 2014] but has not been quantified as a deployment-time RAG signal (additional coherence references appear in the supplementary material). Shi et al. [2023] and Liu et al. [2024] document distraction by irrelevant context and lost-in-the-middle behaviour respectively; we treat the Context Coherence Score (CCS) and answer-bearing evidence presence as two complementary context-structure variables to *report* alongside per-passage similarity, not as substitutes for human evaluation or cross-scorer checks. The supplementary material reproduces a positioning table mapping each prior framework to the audit axis it most directly engages.

3 Setup and Audited Cells

We hold the entire RAG pipeline fixed across audit experiments and vary only the audit axis under test. The audited cells are deliberately narrow—primarily short-answer extractive QA on SQuAD and PubMedQA with 7B-class generators—so that scorer, retriever, and generator choices can be cleanly separated.

Datasets. The matched-context, scaled, metric-fragility, and span-control experiments use SQuAD [Rajpurkar et al., 2016]; the threshold-transfer and noise experiments additionally use PubMedQA [Jin et al., 2019], HotpotQA [Yang et al., 2018], NaturalQuestions [Kwiatkowski et al., 2019], and TriviaQA. The long-form stress test on QASPER and MS-MARCO is supplement-only and not treated as broad generalization evidence (§12 and the supplementary material).

Pipeline. Embeddings come from sentence-transformers/all-MiniLM-L6-v2; indexing uses ChromaDB; refinement uses cross-encoder/ms-marco-MiniLM-L-6-v2. Default chunk size is 1024 characters with 10% overlap and default top- k is 3. Generators are served via local Ollama (Mistral-7B, Qwen2.5-7B). The matched-context audit also re-encodes the same 200 pairs with BGE-large and E5-large.

Conditions. Three retrieval configurations act as instruments for measuring the audit surface, not as systems to crown a winner. We name them by function and (in parentheses) by the prior naming convention: **no-refinement** (top- k dense retrieval, also called *baseline*); **aggressive-refinement** (refine every retrieved passage through sub-chunking and cross-encoder selection; the prior-work name is *HCPC-v1* and this is the original failure-mode condition the published refinement paradox was based on); and **gated-refinement** (a coherence-gated variant with top-rank protection, a refinement cap, and merge-back; prior-work name *HCPC-v2*). Tables and section text use the functional names on first reference and the short prior names thereafter.

Metrics. Per-query **faithfulness** in the legacy audit tables is the stored automated support score in $[0, 1]$; **hallucination** indicates score < 0.5 unless a table states another label. The frozen DeBERTa and roberta-large-mnli scores are sentence-level zero-shot entailment-label proxies (answer sentence classified against entailment / neutral / contradiction with hypothesis_template="{ }"; the retrieved context is retained in the per-row data but not consumed by either proxy). The RAGAS-style judge is context-conditioned by construction: a local Mistral-7B receives the question, retrieved context, and answer and returns a support score on $[0, 1]$. **Retrieval similarity** is the mean cosine similarity between the query and each retained chunk. **CCS** is the mean minus the standard deviation of off-diagonal pairwise cosine similarities between retained chunk embeddings (same embedder). The answer-span indicator is a lowercased, punctuation-stripped lexical match of any normalised gold-answer alias in the retrieved context.

Statistical protocol. The matched-context test is pre-specified with paired Wilcoxon and a 10,000-resample bootstrap CI on the mean paired difference; the scaled audit reports seed-level paired Wilcoxon over five seeds; binary rates use Wilson 95% CIs. P-values outside the pre-specified matched test are descriptive across multiple audit cells; we do not treat exploratory $p < 0.05$ rows as standalone discoveries. All audit cells, paired contrasts, and per-query CSVs are in the anonymized review artifact.

4 ControlledRAG: The Minimum Reporting Standard

ControlledRAG is a checklist. A paper that makes a RAG faithfulness claim fills in seven entries, one per axis, and a reviewer reads those entries to decide whether the claim is generator-universal, scorer-dependent, threshold-dependent, or otherwise narrow. The reporting-checklist template ships with the artifact as controlledrag_reporting_checklist.md (a filled-in copy for this paper is included; this is *distinct* from the mandatory NeurIPS paper checklist that ships at the end of the submitted PDF). Each entry asks: what was held fixed, what was varied, why this choice, and what is the known sensitivity to alternative reasonable values? The checklist is a disclosure layer: one targeted review question per axis.

Figure 1 summarizes the reporting standard as an audit layer over the ordinary RAG evaluation pipeline.

Why seven axes, not six or eight. A movable degree of freedom belongs on the checklist if leaving it unreported lets a reader substitute a different reasonable value and reach a different conclusion. The seven axes the checklist asks every RAG faithfulness paper to report, with the overclaim each blocks, are: (1) *generator* (model family, size, decoding, fixed vs. regenerated

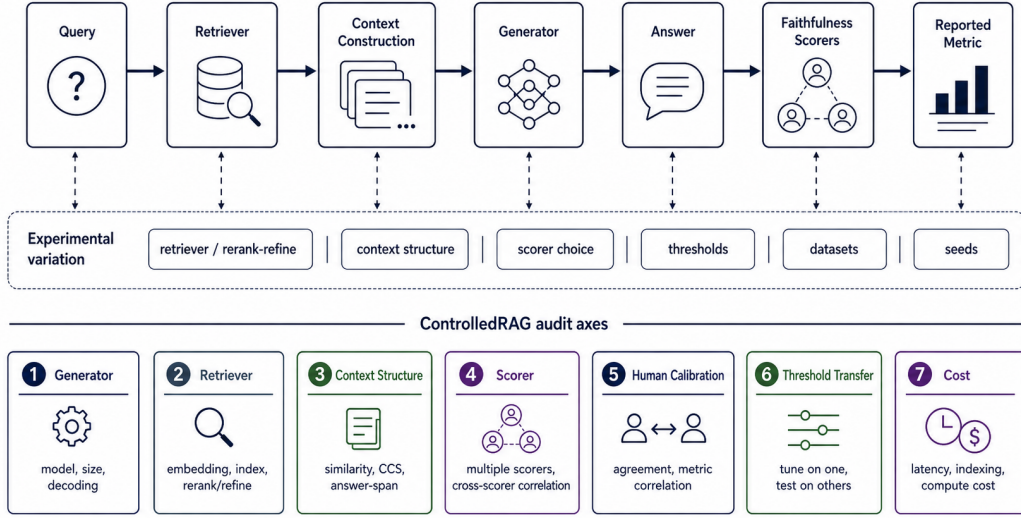


Figure 1: ControlledRAG overview. Faithfulness claims pass through a standard RAG pipeline and are shaped by experimental choices in retrieval, context construction, scoring, thresholding, datasets, and seeds. ControlledRAG asks authors to report seven audit axes so that single-score claims become auditable evidence packages.

136 outputs) blocks generator-universal claims; (2) *retriever* (embedder, index, reranker, refinement
137 policy) blocks “retrieval quality” as one variable; (3) *context structure* (CCS, answer span, mean
138 similarity) blocks CCS-as-oracle; (4) *scorer* (scorer names, input format, thresholds, cross-scorer
139 correlations) blocks single-score claims (§7); (5) *human calibration* (label protocol, raters, agreement,
140 and scorer-alignment correlations on at least one typical-row slice *and* on any targeted disagreement-
141 driven slice) blocks treating any metric ranking as settled (§8); (6) *threshold transfer* (tuned τ , tune
142 dataset, off-diagonal recovery table) blocks reusable-threshold claims (§9); (7) *cost* (per-condition
143 latency, indexing time, external-call assumptions, including baselines) blocks leaderboards without
144 cost (§10). Six axes drop one of these movable quantities; eight start to double-count (decoding is a
145 generator subfield; index size is a retriever subfield) without adding a new degree of freedom. The
146 remainder of the paper supplies a self-applied audit in which each empirical cell motivates one axis.

147 5 The Matched-Context Audit

148 The matched-context audit asks the diagnostic question directly: *at fixed mean per-passage query-*
149 *passage similarity, does higher CCS predict higher faithfulness?* The null answer is not the end of
150 the analysis. It creates a constructive diagnostic question: what variables distinguish answer support
151 once mean similarity is fixed?

152 **Construction.** For SQuAD with Mistral, we form 200 HIGH/LOW pairs so that mean query-
153 passage similarity is matched to within 0.02 (0.4236/0.4215) while CCS is large in HIGH and small
154 in LOW (0.670/0.137).

155 **Result.** Table 1 reports the pre-specified HIGH-LOW contrast. The mean paired difference is
156 -0.002 , paired Wilcoxon $p = 0.628$, Cohen’s $d_z = -0.017$, 10,000-resample bootstrap 95%
157 CI $[-0.022, +0.017]$. The HIGH hallucination rate is also *higher* than the LOW rate (16.5% vs.
158 9.0%). The pre-specified H_1 (HIGH>LOW faithfulness) is not supported in the audited cell. The
159 continuous mean contrast and the thresholded hallucination contrast measure different parts of the
160 score distribution, so a null mean can coexist with threshold-sensitive tail differences. The matched
161 test rules out the simple HIGH-CCS-improves-faithfulness story in this audited cell.

Table 1: Matched-context audit on SQuAD/Mistral ($n=200$ pairs, matched mean query–passage similarity).

Quantity	HIGH-CCS	LOW-CCS	Difference	Test
Mean similarity	0.4236	0.4215	+0.0021	matched by construction
Mean CCS	0.670	0.137	+0.533	—
Faithfulness	0.636	0.639	−0.002	paired Wilcoxon $p = 0.628$
Hallucination	16.5%	9.0%	+7.5 pp	McNemar $p = 0.011$
95% CI on faith diff	—	—	[−0.022, +0.017]	10,000-resample bootstrap

Span presence in matched contexts. A lexical gold-answer-string check finds the answer more often in LOW-CCS contexts than HIGH-CCS contexts (27.0% vs. 17.0%, McNemar $p = 0.011$). The structural property captured by CCS is therefore not the same as “the retrieved context contains the answer”; the lexical span check is a weak proxy because retrieved text can contain the answer string while failing to support the generated answer.

Answer-span/control diagnostic. A regression and classification panel on the matched cell ($n = 400$ contexts; full table in the supplementary material) shows that for continuous DeBERTa faithfulness, answer-span presence is the strongest single predictor (R^2 ranking $0.019 > 0.003 > 0.002$), while for binary hallucination AUROC, CCS is the strongest single predictor ($0.639 > 0.586 > 0.531$). The joint feature set has the highest score on both ($R^2 = 0.025$, AUROC = 0.643). Span and CCS therefore diagnose complementary failure modes; we do not claim a directional role for either, that span subsumes CCS, or that CCS subsumes span.

6 The Scaled Audit

The scaled audit asks the same headline question—does aggressive-refinement (HCPC-v1) reliably hurt faithfulness on SQuAD/Mistral?—at $n = 2,500$ examples per condition with five seeds, each analysed by paired Wilcoxon on 500-pair contrasts. Pooled means are 0.661/0.650/0.661 for no-refinement / aggressive-refinement / gated-refinement with overlapping Wilson CIs on hallucination rate (14.6%/14.7%/14.3%; Table 2). One of five seeds reaches $p < 0.05$ for the no-refinement–aggressive-refinement contrast (seed 44, paired $p = 0.008$); the other four yield $p \in [0.083, 0.234]$. The gated–aggressive contrast reaches significance in 3 of 5 seeds with small per-seed magnitude (0.003–0.019).

Table 2: Scaled audit on SQuAD/Mistral, $n = 2,500$ per condition, pooled across five seeds. Conditions are functional names with prior naming in parentheses.

Condition	Faithfulness (95% CI)	Hallucination (Wilson)	Retrieval sim.	Refine rate
no-refinement (baseline)	0.661 [0.655, 0.667]	14.6% [13.3, 16.0]	0.532	0.000
aggressive-refinement (HCPC-v1)	0.650 [0.645, 0.656]	14.7% [13.4, 16.2]	0.569	1.000
gated-refinement (HCPC-v2)	0.661 [0.655, 0.667]	14.3% [13.0, 15.7]	0.532	0.705

The originally published $n = 30$ headline ($0.7995 \rightarrow 0.6407$ on no-refinement vs. aggressive-refinement; see footnote 1) does not survive at scale. The scaled contrast is small and seed-sensitive and should not be cited as a broad refinement-hurts-faithfulness claim. This is itself a ControlledRAG-style audit lesson: at $n = 30$ the contrast looked clean enough to report as a phenomenon, but at $n = 2,500$ it does not.

7 Fixed-Generation Metric Fragility

For the scaled audit’s 7,500 Mistral rows, generations and retrieval conditions are fixed; only the scorer changes. The scorers are not on a common numerical scale—two are legacy zero-shot label proxies that do not consume retrieved context, and one is a context-conditioned local judge—so raw contrasts are scorer-native rather than directly comparable causal estimates. We therefore also z-score and rank-normalise each scorer over the same fixed rows. The sign, magnitude, and ranking remain scorer-sensitive. A compact Qwen2.5 second-generator probe ($n = 100$ fixed contexts per

condition, one model only) checks whether the raw-contrast ordering is preserved without making a population-level replication claim.

Table 3: Fixed-generation metric fragility. Raw values are no-refinement—aggressive-refinement score differences on each scorer’s native scale and *are not directly comparable across scorers*. Mistral z/rank columns standardize each scorer over the same fixed rows before computing the paired contrast; bootstrap CIs are in the supplementary material. The first two rows are legacy zero-shot label proxies that do not consume the retrieved context (§3); the third is a context-conditioned local judge.

Scorer	Mistral raw	Mistral z	Mistral rank	Qwen2.5 raw
Legacy DeBERTa proxy (no context)	0.011	0.071 SD	0.022	0.015
Legacy roberta-large-mnli proxy (no context)	0.032	0.231 SD	0.062	0.046
Context-conditioned RAGAS-style judge	0.140	0.337 SD	0.085	0.134

Cross-scorer Pearson correlations on the same fixed Mistral rows are weak between the legacy DeBERTa proxy and the other two scorers (DeBERTa vs. roberta-large-mnli: $r = 0.259$; DeBERTa vs. RAGAS-style: $r = 0.182$) and moderate between the roberta-large-mnli proxy and the RAGAS-style judge ($r = 0.674$); the supplementary material shows the same ordering on the Qwen2.5 second-generator probe. The two NLI-style rows are legacy zero-shot entailment-label proxies (answer sentence as the sequence to be classified, retrieved context not consumed); the RAGAS-style row is context-conditioned by construction (the question, retrieved context, and answer are all passed to a local Mistral-7B that returns a support score with a short rationale). Scorer choice and scorer input format are therefore an audit axis, not hidden implementation trivia.

Scorer input format is itself an audit axis. To make the input-format claim concrete, we rescore a balanced subset of the same fixed-generation rows ($n = 200$ paired queries per condition, $n = 600$ total) under a context-conditioned NLI calling convention (premise = retrieved context, hypothesis = generated answer sentence) for both DeBERTa-v3 and roberta-large-mnli. Figure 2 reports the resulting baseline—aggressive-refinement contrasts. Under context-conditioned NLI the same two backbone models produce *sign-flipped* contrasts on the same fixed generations: the legacy and context-conditioned bootstrap intervals are disjoint for both NLI backbones. Passing the retrieved context as premise to the same backbone weights changes the *sign* of the reported contrast at this audit cell.

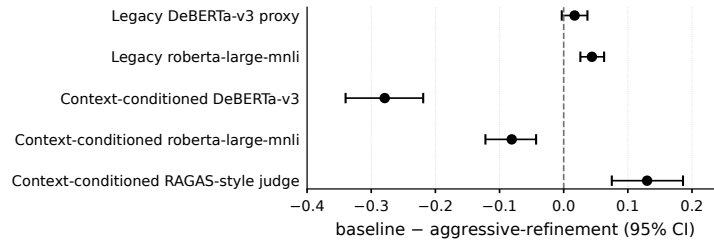


Figure 2: Same fixed generations, same NLI backbones, sign-flipped contrasts under different calling conventions (baseline—aggressive-refinement on the $n = 600$ balanced subset; 10,000-resample paired bootstrap 95% CIs).

Interpretation. Scorer choice is the largest movable quantity in this audit’s headline number, and it remains movable after standardisation. Context-conditioned NLI also flips the sign for both NLI backbones on the same fixed generations. The $n = 99$ adjudicated calibration (§8.1) has overlapping bootstrap CIs across scorers, so it does not settle a ranking. ControlledRAG therefore treats scorer choice, scorer input format, and cross-scorer correlation as reportable audit axes.

8 Human Calibration: Two Slices, Different Scorer Rankings

The human-calibration axis has two completed parts: a **typical-row calibration** ($n = 99$ stratified examples, two passes plus adjudication) and a **targeted disagreement slice** ($n = 100$ examples ranked by scorer disagreement, two raters plus adjudication). The first estimates scorer alignment on

typical rows; the second stress-tests rows where automated scorers most disagree. The two slices reach *different* scorer rankings, making slice-dependence the load-bearing finding.

8.1 Typical-row calibration ($n = 99$).

We sampled 99 rows from the 7,500 Mistral fixed-generation audit, stratified across no-refinement / aggressive-refinement / gated-refinement, and applied a two-pass annotation protocol with three labels (supported, partially_supported, unsupported) plus an adjudication step. Annotators used only the retrieved context. The two passes reach raw exact-match agreement 0.919 (91/99), Cohen’s $\kappa = 0.774$, with adjudicated distribution 79 / 16 / 4 (Table 4). Spearman correlations rank RAGAS-style above the roberta-large-mnli proxy above the legacy DeBERTa proxy, but the three bootstrap 95% CIs overlap, so the per-scorer ordering on this typical slice is suggestive only. The supplementary material contains Pearson and Kendall variants and the per-item verification trace.

8.2 Targeted disagreement slice ($n = 100$).

We construct a separate 100-example scorer-disagreement evaluation where automated scorers disagree most sharply. The batch is *not* a random sample of the 7,500 rows: items are ranked by absolute DeBERTa-proxy/RAGAS-style disagreement, balanced across DeBERTa-high/RAGAS-low, RAGAS-high/DeBERTa-low, near-threshold disagreement, and the three retrieval conditions; items overlapping the $n = 99$ calibration are excluded. Two independent raters labelled all 100 items as faithful / hallucinated / unclear using only the retrieved context, and adjudication resolved the 12 rater-rater disagreements with documented notes.

Table 4: Completed human-evaluation slices: $n = 99$ typical-row calibration vs. $n = 100$ targeted disagreement slice. The two slices answer different audit questions and reach different scorer rankings; that slice-dependence is the load-bearing finding. Bootstrap 95% CIs are paired bootstrap from the verification scripts; full per-quantity CIs are in the supplementary material.

Quantity / scorer	$n = 99$ typical-row calibration		$n = 100$ targeted disagreement	
	Value	95% CI	Value	95% CI
n adjudicated	99	—	100	—
Adjudicated label dist.	79/16/4	supp/part/unsup	74/26/0	faith/hall/unclear
Raw agreement	0.919	91/99	0.88	[0.81, 0.94]
Cohen’s κ	0.774	—	0.721	[0.588, 0.844]
Spearman, DeBERTa proxy	+0.103	[−0.080, +0.283]	−0.156	[−0.331, +0.032]
Spearman, roberta-large-mnli	+0.394	[+0.222, +0.543]	+0.446	[+0.293, +0.581]
Spearman, RAGAS-style	+0.549	[+0.380, +0.707]	+0.384	[+0.241, +0.523]
AUROC, DeBERTa proxy	—	—	0.398	[0.273, 0.514]
AUROC, roberta-large-mnli	—	—	0.794	[0.699, 0.879]
AUROC, RAGAS-style	—	—	0.718	[0.637, 0.793]

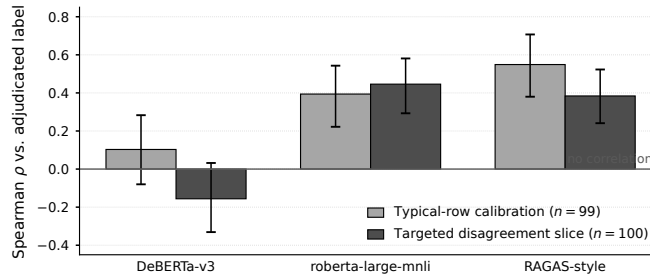


Figure 3: Scorer ranking against adjudicated humans is slice-dependent: RAGAS-style leads on typical rows; roberta-large-mnli leads on the disagreement slice, where the legacy DeBERTa proxy turns weakly negative. Error bars are bootstrap 95% CIs.

Interpretation. Figure 3 shows the inversion: on the targeted disagreement slice roberta-large-mnli is the strongest-aligned scorer (Spearman $\rho = 0.446$, AUROC 0.794,

AUPRC 0.929); RAGAS-style is second ($\rho = 0.384$, AUROC 0.718); the legacy DeBERTa proxy turns weakly negative ($\rho = -0.156$, AUROC 0.398). This is *not* the typical-row ordering. We do not claim any single scorer is universally best, and we do not treat the $n = 100$ slice as a population-level estimate.

9 Threshold Transfer

Recovery is the dimensionless normalised contrast $R_\tau(d) = (\text{faith}_{\text{gate},\tau}(d) - \text{faith}_{v1}(d)) / (\text{faith}_{\text{base}}(d) - \text{faith}_{v1}(d))$; values can exceed 1 or be negative when the no-refinement–aggressive-refinement denominator is small or sign-inverted, so the table is a transport diagnostic, not a universal utility score. We sweep $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ for a CCS-gate retrieval intervention. For each tune dataset, τ^* maximises recovery on that dataset; diagonal recovery evaluates τ^* in-domain, and off-diagonal recovery evaluates it on the other four datasets.

Table 5: Threshold transfer summary. For each tune dataset, τ^* is the value of τ that maximises recovery on that dataset alone; “diagonal recovery” is recovery at τ^* on the same dataset; “off-diagonal mean” is the mean recovery at τ^* over the four *other* eval datasets. A large positive gap means the tuned τ^* does not transport. The full per-cell transfer matrix is in the supplementary material.

τ tuned on	τ^*	Diagonal recovery	Off-diagonal mean	Gap (diag – off-diag)
PubMedQA	0.4	+1.452	+0.451	+1.002
NaturalQs	0.5	+1.401	+0.211	+1.191
SQuAD	0.3	+0.823	−0.141	+0.963
TriviaQA	0.4	+0.464	+0.698	−0.234
HotpotQA	0.3	−0.112	+0.093	−0.205

Tune-SQuAD and tune-HotpotQA both pick $\tau^* = 0.3$, and tune-PubMedQA and tune-TriviaQA both pick $\tau^* = 0.4$, so the off-diagonal columns coincide for those tune-dataset pairs in the full 5×5 matrix (see the supplementary material); this is a faithful consequence of the experiment, not a duplicate-row bug.

Interpretation. PubMedQA, NaturalQuestions, and SQuAD show large diagonal-vs-off-diagonal gaps: transporting the tuned τ loses most of the on-diagonal normalised contrast. The sign-changing cells are the load-bearing threshold-transfer finding.

10 Cost-Aware Baselines

Faithfulness numbers reported without latency and indexing cost are under-identified along the cost axis. We run a head-to-head on $n = 200$ SQuAD and $n = 200$ HotpotQA queries with CRAG [Yan et al., 2024], gated-refinement (HCPC-v2), and a 2-level RAPTOR [Sarathi et al., 2024] index.

Table 6: Cost-aware head-to-head, $n = 200$ per (dataset, condition). Latency is end-to-end per query; indexing time is the offline cost of building the dataset’s retrieval structure. Faithfulness CIs are bootstrap 95% intervals; hallucination CIs are Wilson 95% intervals. RAPTOR-2L’s indexing cost is $17\text{--}22\times$ the dense-baseline ($n = 200$ per cell).

Dataset	Condition	Faithfulness	Hallucination	Mean latency	Indexing
SQuAD	CRAG	0.698 [0.675, 0.722]	12.0% [8.2, 17.2]	1,440 ms	8.0 s
SQuAD	gated-refinement (HCPC-v2)	0.708 [0.685, 0.732]	12.5% [8.6, 17.8]	1,720 ms	8.0 s
SQuAD	RAPTOR-2L	0.710 [0.685, 0.735]	12.5% [8.6, 17.8]	1,714 ms	175.4 s
HotpotQA	CRAG	0.643 [0.627, 0.658]	10.5% [7.0, 15.5]	1,941 ms	8.0 s
HotpotQA	gated-refinement (HCPC-v2)	0.633 [0.617, 0.650]	12.5% [8.6, 17.8]	2,289 ms	8.0 s
HotpotQA	RAPTOR-2L	0.631 [0.615, 0.647]	13.0% [9.0, 18.4]	2,414 ms	134.0 s

Harness and cost accounting. This is a closed, zero-dollar local/free-tier harness: local Ollama/Mistral generation, Chroma indexing, MiniLM cross-encoders, no paid APIs. The CRAG

reimplementation uses the documented evaluator/strip-refinement structure with live web search disabled, so its external-call count is zero. A live-web CRAG would need to report web-call count, API cost, and network latency separately. Self-RAG [Asai et al., 2024] is reported only in the supplementary material because its reflective decoding violates the fixed-context assumption.

At $n = 200$ per cell, faithfulness CIs overlap. The clear separation is indexing cost: RAPTOR-2L is $17\text{--}22\times$ the dense baseline. The audited cell therefore supports a cost tradeoff, not a universal faithfulness winner.

11 Discussion

What survives, by axis. Strength tags reflect evidence weight in this audit, not effect size.

- *Generator* [**WEAK**]: second-generator probe (Qwen2.5, $n = 100$, one model) preserves the metric-fragility ordering; not a panel claim.
- *Retriever* [**MEDIUM**]: matched HIGH>LOW CCS preserved across MiniLM/BGE/E5 in the supplementary material; fixed generations unchanged.
- *Context structure* [**STRONG**]: matched CCS intervention null (-0.002 , paired Wilcoxon $p = 0.628$); span and CCS are complementary diagnostic axes.
- *Scorer* [**STRONG**]: raw contrasts $0.011/0.032/0.140$ on the same fixed generations; context-conditioned NLI sign-flips both backbones with disjoint bootstrap CIs (§7).
- *Human calibration* [**STRONG**]: $n = 99$ ($\kappa = 0.774$) ranks RAGAS-style first; $n = 100$ ($\kappa = 0.721$) ranks roberta-large-mnli first and the legacy DeBERTa proxy weakly negative. Slice-dependent.
- *Threshold* [**MEDIUM**]: 5×5 transfer table has sign-changing cells; no bootstrap CIs.
- *Cost* [**WEAK**]: faithfulness CIs overlap at $n = 200$; the indexing-cost gap ($17\text{--}22\times$ for RAPTOR-2L) is the clear signal.

Coherence-preserving and random-noise slopes differ in point estimate, but no significance test was run; see the supplementary material for details. Scope limits on the empirical demonstration are categorised in §12.

Takeaway. All seven axes are load-bearing here, but in different proportions. ControlledRAG asks for each one because leaving any movable quantity unreported can change the headline.

12 Limitations and Ethics

Audited surface. Audited cells are mostly short-answer extractive QA on SQuAD with 7B-class generators. The matched-context audit is SQuAD-only; the long-form QASPER/MS-MARCO probe has 40 distinct questions and is supplement-only; and Self-RAG [Asai et al., 2024] is supplement-only because its harness mismatches the fixed-context protocol. Stronger embeddings preserve the matched HIGH>LOW CCS ordering, but the fixed-generation null is unchanged.

Human evaluation. The typical-row calibration ($n = 99$) has overlapping bootstrap CIs across scorers; the targeted disagreement slice ($n = 100$) is adversarial-by-construction. Both use two raters plus adjudication, not a broad annotator pool, and neither settles a global scorer order.

Scope of replication. The second-generator probe is one local Qwen2.5 model at $n = 100$ fixed contexts per condition (a sanity check, not a panel replication); the context-conditioned NLI rescoring is on a $n = 600$ balanced subset, not the full 7,500 (sufficient to demonstrate input-format sign-flipping, not a full-table replacement).

Ethics. Public datasets and free-tier or local compute only; the annotation arms cover short-answer support judgments on public content (no PII, no sensitive content); annotator identities are not stored. Review artifact anonymized; public links follow acceptance.

These limits narrow the empirical surface but reinforce the methodological point: such scope conditions should be disclosed rather than hidden behind a single score.

13 Conclusion

RAG faithfulness claims should not rest on one retriever, one generator, one context construction, one scorer, one threshold, or one baseline row. Here, the matched CCS intervention is null once mean similarity is controlled; the scaled refinement-paradox contrast is small and seed-sensitive; fixed generations change their reported effect size and even sign under scorer and scorer-input choices; human alignment is slice-dependent; thresholds do not transport cleanly; and faithfulness-only comparisons miss cost. ControlledRAG turns these failure modes into a minimum seven-axis evidence standard for scoped, reproducible faithfulness claims.

References

- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAS: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–163, 2024.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2024.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. TRUE: Re-evaluating factual consistency evaluation. *NAACL*, 2022.
- Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 2567–2577, 2019.
- Vladimir Karpukhin, Barlas Öguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergei Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781, 2020.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.
- Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. In *Transactions of the Association for Computational Linguistics (TACL)*, 2022.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.
- Jiwei Li and Eduard Hovy. A model of coherence based on distributed sentence representation. *Proceedings of EMNLP*, 2014.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics (TACL)*, 2024.
- Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proceedings of the 62nd Annual Meeting of the Association*

for *Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.585. URL <https://aclanthology.org/2024.acl-long.585/>.

Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.

Chanhee Park, Hyeonseok Moon, Chanjun Park, and Heuseok Lim. MIRAGE: A metric-intensive benchmark for retrieval-augmented generation evaluation. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 2883–2900, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-naacl.157. URL <https://aclanthology.org/2025.findings-naacl.157/>.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2383–2392, 2016.

Dongyu Ru, Lin Qiu, Xiangkun Hu, Tianhang Zhang, Peng Shi, Shuaichen Chang, Jiayang Cheng, Cunxiang Wang, Shichao Sun, Huanyu Li, Zizhao Zhang, Binjie Wang, Jiarong Jiang, Tong He, Zhiguo Wang, Pengfei Liu, Yue Zhang, and Zheng Zhang. RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation. *arXiv preprint arXiv:2408.08067*, 2024. doi: 10.48550/arXiv.2408.08067. URL <https://arxiv.org/abs/2408.08067>.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 338–354, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.20. URL <https://aclanthology.org/2024.naacl-long.20/>.

Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2401.18059>.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2023.

Manveer Singh Tamber, Forrest Sheng Bao, Chenyu Xu, Ge Luo, Suleman Kazi, Minseok Bae, Miaoran Li, Ofer Mendelevitch, Renyi Qu, and Jimmy Lin. Benchmarking LLM faithfulness in RAG with evolving leaderboards. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 799–811, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-industry.54. URL <https://aclanthology.org/2025.emnlp-industry.54/>.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Improving text embeddings with large language models. *arXiv preprint arXiv:2401.00368*, 2024.

Fangyuan Xu, Weijia Shi, and Eunsol Choi. RECOMP: Improving retrieval-augmented LMs with compression and selective augmentation. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2310.04408>.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 2564–2579, 2024.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, 2018.

Supplementary Material

A Pre-Specification

The matched-context, scaled, metric-fragility, threshold-transfer, noise, cost-aware-baseline, and human-calibration audits were pre-specified on 2026-04-26 before any results were inspected. The human-calibration design specified two arms: (i) a stratified $n \approx 99$ two-pass adjudicated calibration on the typical row distribution of the 7,500 Mistral generations, scored as raw inter-pass agreement, Cohen’s κ , and Spearman correlations of each scorer against the adjudicated label; (ii) a targeted 100-row scorer-disagreement batch constructed from the same 7,500 rows and designed to stress-test scorer conflict using two independent raters plus an adjudication step. Both arms are now completed: §L reports arm (i) at $n = 99$ adjudicated items, and §M reports arm (ii) at $n = 100$ two-rater adjudicated items. The pre-specification also fixed: the matched-pair construction (similarity gap ≤ 0.02 , CCS gap ≥ 0.20); the paired Wilcoxon test direction (HIGH>LOW); the 10,000-resample bootstrap for the CI on the mean paired difference; the Wilson 95% CI on hallucination rates; the seed list {41, 42, 43, 44, 45}; the per-seed paired Wilcoxon as the primary significance test in the scaled audit; the three faithfulness scorers (DeBERTa proxy, roberta-large-mnli proxy, RAGAS-style) plus the fixed-context rescore protocol; the τ -sweep range; and the 200-query head-to-head sample size. The four follow-up arms were added after the pre-specified runs as post-hoc robustness checks: answer-span/control diagnostic, stronger-retriever sanity check, Qwen2.5 second-generator replication, and long-form stress test. We disclose which experiments are pre-specified (audit_log.md sections under “Pre-Specified”) versus follow-ups (audit_log.md sections under “Follow-Up”) in the file audit_log.md shipped at the top level of the artifact.

B Scaled Audit: Per-Seed Detail

The scaled audit on SQuAD/Mistral runs $n = 2,500$ examples per condition for each of five seeds; pooled means appear in the main paper. Table 7 reports per-seed paired Wilcoxon p -values, mean paired differences, and Cohen’s d_z for each contrast.

Table 7: Per-seed paired contrasts on SQuAD/Mistral, $n = 500$ pairs per seed. Significance is paired Wilcoxon at $\alpha = 0.05$.

Seed	Contrast	Mean diff	Wilcoxon p	Cohen’s d_z
41	baseline—v1	+0.0107	0.083	+0.072
42	baseline—v1	+0.0063	0.234	+0.043
43	baseline—v1	+0.0078	0.174	+0.055
44	baseline—v1	+0.0204	0.008	+0.134
45	baseline—v1	+0.0073	0.203	+0.048
41	v2—v1	+0.0132	0.025	+0.088
42	v2—v1	+0.0030	0.504	+0.021
43	v2—v1	+0.0059	0.215	+0.042
44	v2—v1	+0.0191	0.008	+0.126
45	v2—v1	+0.0125	0.036	+0.081

Bootstrap CIs on the pooled mean paired difference for baseline—v1 include zero, and only 1 of 5 seeds reaches significance for that contrast. The v2—v1 contrast reaches significance in 3 of 5 seeds, but the magnitude is small (per-seed mean diff 0.003–0.019). Seed-level uncertainty is treated as a ControlledRAG reporting axis, not a result to average away.

C Answer-Span/Control Diagnostic Details

The answer-span detector lowercases text, strips punctuation, normalises whitespace, and checks whether any normalised gold-answer alias appears in the retrieved context for the audited RAG setting [Lewis et al., 2020]. It uses no external model. The resulting control rows and model summaries are written under the Fix 12 data and results directories.

Table 8: Matched-context span and outcome summary.

Group	n	span present	faithfulness	hallucination	CCS
all	400	0.220	0.637	0.128	0.404
HIGH-CCS	200	0.170	0.636	0.165	0.670
LOW-CCS	200	0.270	0.639	0.090	0.137
span absent	312	0.000	0.628	0.119	0.408
span present	88	1.000	0.670	0.159	0.390

Table 9: Full span/control model comparison. CIs are bootstrap 95% intervals from the script output.

Predictors	R^2	R^2 CI	AUROC	AUROC CI
answer span	0.019	[0.001, 0.062]	0.531	[0.502, 0.599]
mean similarity	0.002	[0.000, 0.023]	0.586	[0.495, 0.677]
CCS	0.003	[0.000, 0.022]	0.639	[0.564, 0.712]
span + similarity	0.024	[0.004, 0.071]	0.590	[0.518, 0.687]
span + CCS	0.022	[0.003, 0.068]	0.631	[0.572, 0.716]
span + similarity + CCS	0.025	[0.006, 0.075]	0.643	[0.589, 0.734]

447 The headline reading is reported in the main paper: span presence is the strongest single predictor of
 448 continuous faithfulness, CCS is the strongest single predictor of hallucination AUROC, and the joint
 449 feature set has the highest score on both. The R^2 values are uniformly small in absolute terms; the
 450 relative ranking is the load-bearing signal. Span presence is a weak lexical proxy, not a gold support
 451 variable: retrieved text can contain the answer string while still failing to support the generated
 452 answer. The mean-score and thresholded hallucination views also inspect different parts of the score
 453 distribution.

Table 10: Matched HIGH/LOW CCS faithfulness distribution.

Group	n	mean	p10	p25	p50	p75	p90
HIGH-CCS	200	0.636	0.476	0.553	0.638	0.713	0.795
LOW-CCS	200	0.639	0.512	0.569	0.648	0.715	0.769

Table 11: Matched HIGH/LOW CCS hallucination rates as the score threshold varies.

Group	$\theta = 0.3$	$\theta = 0.4$	$\theta = 0.5$	$\theta = 0.6$	$\theta = 0.7$
HIGH-CCS	0.0%	3.0%	16.5%	38.5%	71.0%
LOW-CCS	1.0%	3.0%	9.0%	38.5%	70.0%

454 These tables explain why the matched mean contrast can be null while a specific thresholded
 455 hallucination contrast differs: the distributions are close in the middle, and the observed HIGH/LOW
 456 difference is threshold-sensitive rather than a clean directional CCS effect.

457 D Stronger Retriever Sanity Check

458 The retriever sanity check has two parts. First, the matched contexts from the main-paper audit
 459 are re-encoded with MiniLM, BGE-large, and E5-large; this is a fixed-context diagnostic, not
 460 a fresh generation run. Second, the completed multi-retriever generation artifact is copied into
 461 data/revision/fix_13/ and summarised. The fixed generations are unchanged across embedders,
 462 so the matched HIGH—LOW faithfulness contrast inherits the matched-context null from the main
 463 paper and is not re-tested here.

464 The safe interpretation is retriever sensitivity, not universal robustness. Stronger embeddings change
 465 the retrieval surface, sometimes in a direction that flips the SQuAD paradox sign. ControlledRAG
 466 asks for retriever identity to be reported as a routine audit axis; this appendix section is the operational
 467 example.

Table 12: Fixed-context re-encoding of the matched HIGH/LOW CCS pairs. The HIGH>LOW CCS ordering is preserved across all three embedders.

Embedder	pairs	CCS high	CCS low	high-low	sim gap
MiniLM	200	0.670	0.137	0.533	0.006
BGE-large	200	0.739	0.491	0.249	0.032
E5-large	200	0.830	0.702	0.127	0.015

Table 13: Paradox magnitude by embedding model from the completed multi-retriever generation artifact ($n = 30$ per cell, original pilot sampling protocol re-run under different embedders). The SQuAD column shows that the paradox direction depends on retriever; PubMedQA preserves a positive paradox across all four embedders. *These rows are not directly comparable to the $n = 2,500$ scaled-audit number in main paper §6 (baseline-v1 = 0.011): that table uses the audit-construction sampling and a single MiniLM retriever, whereas this table uses the small- n pilot sampling and varies the embedder. Cross-table comparison should be same-row across embedders within this table, not this table vs. the main-paper scaled-audit table.*

Dataset	embedder	baseline-v1	v2-v1
SQuAD	MiniLM	+0.075	+0.099
SQuAD	BGE-large	+0.032	+0.025
SQuAD	E5-large	-0.011	+0.008
SQuAD	GTE-large	-0.001	-0.006
PubMedQA	MiniLM	+0.024	+0.019
PubMedQA	BGE-large	+0.033	+0.028
PubMedQA	E5-large	+0.053	+0.039
PubMedQA	GTE-large	+0.006	+0.002

E Second-Generator Probe (Qwen2.5)

The second-generator audit samples $n = 100$ fixed contexts per condition from the matched-context construction and generates answers with Qwen2.5 via local Ollama. Those fixed generations are scored with the DeBERTa proxy, the roberta-large-mnli proxy, and a local RAGAS-style judge (Mistral via Ollama).

Table 14: Qwen2.5 fixed-generation metric means and raw score contrasts.

Metric	baseline	HCPC-v1	HCPC-v2	baseline-v1	v2-v1
DeBERTa	0.645	0.631	0.644	+0.015	+0.014
roberta-large-MNLI	0.362	0.316	0.368	+0.046	+0.051
RAGAS-style	0.806	0.672	0.814	+0.134	+0.142

Table 15: Metric correlations on the Qwen2.5 fixed-generation rows ($n = 300$).

Metric pair	Pearson r	Spearman ρ
DeBERTa vs. roberta-large-MNLI	0.380	0.429
DeBERTa vs. RAGAS-style	0.280	0.318
roberta-large-MNLI vs. RAGAS-style	0.721	0.668

The scorer-disagreement ordering is preserved under this compact second-generator replication. The claim is limited to this audited Qwen2.5 cell; ControlledRAG explicitly asks for a broader generator panel before any universal-generator claim is made.

F Scorer Implementation and Standardized Contrasts

The frozen artifact uses two legacy NLI-style zero-shot label proxies and one context-conditioned RAGAS-style judge. The DeBERTa proxy (cross-encoder/nli-deberta-v3-base) and RoBERTa proxy (roberta-large-mnli) score each generated answer sentence as the sequence to

be classified against the labels entailment/neutral/contradiction with `hypothesis_template="{}`". The retrieved context is retained in the per-row data and used by the RAGAS-style judge, but it is not consumed by those two legacy zero-shot proxy calls. This disclosure is part of the scorer audit: model identity, input formatting, and prompt format are not implementation trivia.

Because the three scorers are on different native scales, the main paper reports raw contrasts as score-scale-dependent contrasts and adds a standardized Mistral analysis. Table 16 z-scores each scorer over the same 7,500 fixed-generation rows and also reports percentile-rank-normalized contrasts. CIs are paired bootstrap intervals over matched query/seed keys; 2,499 paired keys are available after one duplicated key is collapsed by the pivot.

Table 16: Standardized fixed-generation scorer-fragility contrasts on Mistral rows. Positive values mean the left condition scores higher than HCPC-v1.

Scorer	Contrast	raw diff	z diff (CI)	rank diff (CI)
DeBERTa proxy	baseline—v1	0.010	0.071 [0.032, 0.110]	0.022 [0.010, 0.033]
DeBERTa proxy	v2—v1	0.011	0.073 [0.034, 0.112]	0.022 [0.011, 0.034]
roberta-large-MNLI proxy	baseline—v1	0.032	0.231 [0.191, 0.271]	0.062 [0.051, 0.074]
roberta-large-MNLI proxy	v2—v1	0.032	0.236 [0.199, 0.275]	0.063 [0.052, 0.075]
RAGAS-style judge	baseline—v1	0.139	0.337 [0.298, 0.375]	0.085 [0.075, 0.095]
RAGAS-style judge	v2—v1	0.138	0.332 [0.294, 0.370]	0.083 [0.073, 0.092]

The standardized table preserves the scorer-fragility conclusion without treating raw scores on different scales as directly comparable causal effects.

G Context-Conditioned NLI Rescoring

The frozen artifact’s DeBERTa-v3 and roberta-large-mnli faithfulness columns are legacy zero-shot label proxies: the answer sentence is the sequence classified against the labels entailment / neutral / contradiction with `hypothesis_template="{}`", and the retrieved context is not consumed by those calls. To make scorer input format an empirically observable audit axis (not just a disclosure axis), we rescore a balanced subset of the same fixed-generation rows under a proper context-conditioned NLI calling convention:

premise = retrieved context (truncated to 2048 chars), hypothesis = generated answer sentence,

followed by softmax over {contradiction, neutral, entailment} and a mean-over-sentences aggregation matching the legacy proxy aggregation. The rescoring script is `scripts/run_context_conditioned_nli_scoring.py`.

Subset. $n = 200$ paired queries per condition (baseline, HCPC-v1, HCPC-v2), so $n = 600$ rows total. Subset selection is deterministic (first 200 rows by appearance under the original `fix_03` ordering), so the subset is balanced across conditions but is not stratified by question domain. Rescoring is performed with both `cross-encoder/nli-deberta-v3-base` and `roberta-large-mnli` on Apple-silicon MPS without paid APIs.

Table 17: Baseline—HCPC-v1 paired contrasts on the same fixed generations under five scorer input formats. Bootstrap 95% CIs are 10,000-resample paired bootstraps over matched query keys. The two legacy zero-shot proxies (rows 1–2) and the two corresponding context-conditioned NLI rescoring (rows 3–4) use the same backbone NLI weights and the same fixed answers; only the calling convention changes.

Scorer	Baseline—HCPC-v1 contrast	95% CI
Legacy DeBERTa-v3 proxy (no context)	+0.017	[−0.003, +0.037]
Legacy roberta-large-mnli proxy (no context)	+0.044	[+0.026, +0.063]
Context-conditioned DeBERTa-v3 NLI	−0.279	[−0.340, −0.219]
Context-conditioned roberta-large-mnli NLI	−0.081	[−0.122, −0.043]
Context-conditioned RAGAS-style judge	+0.130	[+0.075, +0.186]

Reading. For both DeBERTa-v3 and roberta-large-mnli, switching from the legacy zero-shot label proxy to a context-conditioned NLI calling convention on the same backbone weights changes the sign of the reported baseline—HCPC-v1 contrast on the same fixed generations. The legacy and context-conditioned bootstrap CIs are disjoint for both models. The RAGAS-style judge, which is context-conditioned by construction, agrees in sign with the legacy proxies but disagrees in magnitude. We do not claim a uniformly “correct” scorer; the table is the audit point: scorer input format is a real movable quantity in the headline number.

Alignment with the $n = 99$ calibration. On the $n = 99$ adjudicated calibration items, switching the same DeBERTa-v3 and roberta-large-mnli backbones to the context-conditioned calling convention does *not* improve Spearman alignment with the adjudicated labels: legacy DeBERTa proxy $\rho = +0.157$ vs. context-conditioned DeBERTa $\rho = +0.120$; legacy roberta-large-mnli proxy $\rho = +0.393$ vs. context-conditioned $\rho = +0.224$ (CIs in n99_alignment_spearmans.csv). The result is consistent with NLI models pre-trained on short premise/hypothesis pairs being sensitive to long-context input format. This is a calibration finding, not a recommendation; the contribution is that scorer input format must be reported and audited, not that any one input format is preferred.

Subset is not the full table. This is a 600-row subset, not a full 7,500-row rescoring. The scaled audit raw table in the main paper still uses the full 7,500-row legacy proxy values; the standardised z-/rank-contrast table (§F) also uses the legacy proxies. The context-conditioned subset is a focused empirical demonstration of the input-format axis, not a replacement for the legacy table.

Source files. scripts/run_context_conditioned_nli_scoring.py, results/revision/context_conditioned_nli/per_query_n600.csv, results/revision/context_conditioned_nli/contrasts_n600.csv, results/revision/context_conditioned_nli/n99_with_ctx_scores.csv, results/revision/context_conditioned_nli/n99_alignment_spearmans.csv.

H Long-Form/Synthesis Stress Test

The long-form stress test packages an existing QASPER/MS-MARCO run with 40 distinct long-form questions. It is included only as an exploratory scope probe, not as broad generalization evidence for the short-answer audit.

Table 18: Long-form stress-test condition means.

Dataset	condition	span faith	claim faith	unsupported	long halluc.
MS-MARCO	baseline	0.645	0.646	0.224	0.400
MS-MARCO	HCPC-v1	0.678	0.679	0.075	0.050
MS-MARCO	HCPC-v2	0.674	0.674	0.158	0.250
QASPER	baseline	0.649	0.649	0.158	0.300
QASPER	HCPC-v1	0.647	0.647	0.117	0.200
QASPER	HCPC-v2	0.648	0.651	0.129	0.250

The short-answer paradox does not generalise cleanly here: MS-MARCO has a negative span-faithfulness paradox drop (-0.033), and QASPER is near zero ($+0.002$). The long-form result is reported as a scope limit rather than as broad long-form evidence for ControlledRAG.

I Self-RAG: Harness-Mismatched Appendix Baseline

We include Self-RAG [Asai et al., 2024] only as an appendix baseline because the available Self-RAG harness mismatches the audit’s fixed-context protocol along three dimensions: (i) Self-RAG’s generator emits its own retrieval and reflection tokens, so the “fixed retrieved context” assumption used by the matched-context and metric-fragility audits is not directly satisfied; (ii) Self-RAG’s recommended generator is a fine-tuned Llama-2-7B variant that does not coincide with the audited Mistral-7B / Qwen2.5-7B family and that was OOM-killed on the available T4 instance for the PubMedQA cell, so the partial SQuAD numbers we have are not directly comparable to the cost-aware head-to-head; and (iii) the recommended Self-RAG threshold differs from the threshold-transfer

sweep used elsewhere. We report this disclosure here so that future ControlledRAG-style audits can budget for a matched Self-RAG harness; we explicitly do not generalize from these mismatched runs to Self-RAG overall.

J Coherence vs. Generic Retrieval Noise

The noise audit uses the matched-context construction as a backbone and replaces 1/3, 2/3, or 3/3 of the retrieved chunks with one of two noise types: random off-topic passages and coherence-preserving uninformative passages drawn from the same topic distribution that do not contain the answer.

Table 19: Faithfulness and similarity slopes per noise rate.

Condition	faith slope (per noise rate)	sim slope (per noise rate)
random noise	−0.069	−0.481
coherent uninformative noise	−0.043	−0.113
HCPC-v1 refinement (reference)	—	—

The two noise types produce distinguishable faithfulness slopes at matched substitution rate; we do not claim the gap is large in absolute terms. The right reading is that coherence and similarity are correlated but not identical.

K Threshold Transfer: Full Sweep

Naming. Tables in this section use the prior-work names HCPC-v1 (= aggressive-refinement) and HCPC-v2 (= gated-refinement); see main paper §3.

The main paper reports a per-tune-dataset summary (diagonal recovery, off-diagonal mean, gap) at each tune dataset’s selected τ^* . The full 5×5 per-cell transfer matrix at each tune dataset’s τ^* is reproduced below; the per- τ sweep over $\{0.3, 0.4, 0.5, 0.6, 0.7\}$ for each dataset is in the artifact’s `results/revision/fix_04/` directory. Recovery is the dimensionless normalized contrast

$$R_\tau(d) = \frac{\text{faith}_{\text{gate},\tau}(d) - \text{faith}_{\text{HCPC-v1}}(d)}{\text{faith}_{\text{baseline}}(d) - \text{faith}_{\text{HCPC-v1}}(d)}.$$

It is useful for asking whether a tuned threshold transports, but it is not a direct utility measure: when the denominator is small or negative, the sign can be counterintuitive and values can exceed 1.

Table 20: Full 5×5 threshold-transfer matrix. Each row gives the normalized recovery ratio of the τ tuned on the row dataset, evaluated on each of the five column datasets. Positive values are not always “better” when the baseline-v1 denominator is negative. Rows can be byte-identical across the eval columns when two tune datasets pick the same τ^* : tune-SQuAD and tune-HotpotQA both pick $\tau^* = 0.3$, and tune-PubMedQA and tune-TriviaQA both pick $\tau^* = 0.4$, so the off-diagonal columns coincide. This is a faithful consequence of the experiment (off-diagonal recovery only depends on the eval dataset and the selected τ , not on which tune dataset selected it), not a pipeline bug.

τ tuned on	τ^*	SQuAD	PubMedQA	HotpotQA	NaturalQs	TriviaQA
PubMedQA	0.4	+0.489	+1.452	−0.291	+1.140	+0.464
NaturalQs	0.5	+0.327	+0.664	−0.471	+1.401	+0.323
SQuAD	0.3	+0.823	−1.714	−0.112	+0.865	+0.398
TriviaQA	0.4	+0.489	+1.452	−0.291	+1.140	+0.464
HotpotQA	0.3	+0.823	−1.714	−0.112	+0.865	+0.398

Per- τ recovery curves and confidence diagnostics for the τ -tuning step are included as machine-readable CSVs in the same results directory. The transfer matrix itself does not include bootstrap CIs, so we use it as a descriptive transport diagnostic.

L Completed Annotation Calibration ($n = 99$)

The $n = 99$ annotation calibration set is the typical-row arm of the human-calibration pre-specification. Items were sampled from the 7,500 Mistral fixed-generation rows so that the baseline,

571 HCPC-v1, and HCPC-v2 conditions are each represented and so that the score range of each scorer
572 is covered. Two independent annotation passes were performed using the three-class label scheme
573 supported / partially_supported / unsupported; disagreements were adjudicated and the
574 adjudicated label set is the final reference for scorer-alignment computations.

575 **Adjudicated label distribution.** The adjudicated label distribution is 79 supported, 16
576 partially_supported, 4 unsupported (totals 99). Raw inter-pass agreement on the three-class
577 label is 0.919 (91 of 99 items exact-match), and Cohen’s $\kappa = 0.774$.

Table 21: Annotation calibration agreement and per-scorer alignment on $n = 99$ adjudicated examples. Spearman, Pearson, and Kendall correlations are between the ordered adjudicated label (unsupported < partially_supported < supported) and each scorer’s continuous output on the same items. Bootstrap 95% CIs are paired bootstrap intervals; the three Spearman CIs overlap, so the per-scorer ranking is suggestive only.

Scorer / quantity	Spearman ρ	Pearson r	Kendall τ_b	Spearman CI
Legacy DeBERTa proxy	0.103	0.110	0.083	$[-0.080, +0.283]$
Legacy roberta-large-mnli proxy	0.394	0.404	0.325	$[+0.222, +0.543]$
Context-conditioned RAGAS-style judge	0.549	0.450	0.527	$[+0.380, +0.707]$
Cohen’s κ between the two annotation passes: 0.774				
Raw exact-match agreement: 0.919 (91 of 99)				

578 **Reading.** The Spearman ordering matches the metric-fragility ordering in the main paper (RAGAS-
579 style judge > roberta-large-mnli proxy > DeBERTa proxy), but two facts keep this calibration
580 short of a settled scorer endorsement: (i) the three Spearman bootstrap CIs overlap, and (ii) $n = 99$
581 is calibration sample size, not population sample size. The legacy DeBERTa proxy’s CI overlaps
582 zero, while the other two CIs do not—this is the strongest single line that can be drawn under this
583 calibration.

584 **Source files.** The verification script `scripts/verify_human_eval.py` re-
585 computes the table values from the combined two-pass annotation file
586 `human_eval_final/n99_calibration/human_eval_adjudicated.csv` (carrying
587 `rater_a_label`, `rater_b_label`, `adjudicated_label`, `adjudication_notes`) and
588 emits the bootstrap-CI summary above. The label distribution, agreement sum-
589 mary, correlation table, bootstrap intervals, verification trace, and per-row file with
590 retrieved-context column appear in the same `human_eval_final/n99_calibration/`
591 directory under `human_eval_label_distribution.csv`, `human_eval_summary.csv`,
592 `human_eval_correlations.csv`, `bootstrap_correlation_cis.csv`,
593 `human_eval_verification.csv`, and `human_eval_n99_with_context.csv`.

594 M Completed Targeted Scorer-Disagreement Evaluation ($n = 100$)

595 The targeted scorer-disagreement evaluation is the adversarial arm of the human-calibration pre-
596 specification. It is *separate* from the $n = 99$ typical-row calibration in §L: items are constructed
597 to maximise scorer disagreement rather than to estimate population-level faithfulness. Both raters
598 labelled all 100 items and adjudication resolved the 12 rater-rater disagreements.

599 **Batch construction.** Rows are sampled from the 7,500 Mistral generations in the scaled audit. We
600 compute the absolute disagreement between the legacy DeBERTa proxy and the context-conditioned
601 RAGAS-style score and stratify the batch across three disagreement types:

- 602 1. **DeBERTa high / RAGAS-style low.**
- 603 2. **RAGAS-style high / DeBERTa low.**
- 604 3. **Near-threshold disagreement** around the 0.5 decision boundary.

605 The batch is also balanced across baseline / HCPC-v1 / HCPC-v2. Items already used in the $n = 99$
606 calibration are excluded.

Annotation protocol. Two independent raters assigned one of faithful, hallucinated, or unclear using only the retrieved context, and an adjudication step resolved disagreements. All protocol artifacts below live under `human_eval_final/n100_disagreement/`:

- `annotation_batch_disagreement_100.csv`: merged batch with scorer scores, both rater labels, and adjudicated labels.
- `human_eval_rater1.csv` and `human_eval_rater2.csv`: independent rater files.
- `human_eval_adjudicated.csv`: per-row adjudicated labels, confidence, and notes; all 12 rater disagreements carry meaningful adjudication notes.
- `human_eval_adjudication_template.csv`: the reconciliation template.

Agreement and label distribution. The two raters labelled all 100 items (100/100 for each rater); the adjudicated label set covers all 100. Raw inter-rater agreement is 0.88 (88 of 100 exact-match), Cohen’s $\kappa = 0.721$. The rater 1 distribution is 74 faithful / 26 hallucinated / 0 unclear; the rater 2 distribution is 70 / 20 / 10 (rater 2 used the unclear option, rater 1 did not). The adjudicated distribution is 74 faithful / 26 hallucinated / 0 unclear.

Scorer alignment with adjudicated labels. Table 22 reports per-scorer correlation, AUROC, and AUPRC against the adjudicated faithful / hallucinated label, with 1,000-resample bootstrap 95% CIs from `scripts/analyze_human_disagreement_labels.py`.

Table 22: Two-rater adjudicated $n = 100$ targeted scorer-disagreement evaluation. The slice is constructed to be adversarial for any single scorer; the ranking here is *not* the typical-row ranking from the $n = 99$ calibration. Raw inter-rater agreement is 0.88; Cohen’s $\kappa = 0.721$. The adjudicated distribution is 74 faithful / 26 hallucinated / 0 unclear.

Scorer	Spearman ρ	Pearson r	AUROC	AUROC CI	AUPRC	AUPRC CI
Legacy DeBERTa proxy	-0.156	-0.188	0.398	[0.273, 0.514]	0.765	[0.670, 0.852]
Legacy roberta-large-mnli proxy	+0.446	+0.380	0.794	[0.699, 0.879]	0.929	[0.881, 0.963]
Context-conditioned RAGAS-style judge	+0.384	+0.366	0.718	[0.637, 0.793]	0.859	[0.793, 0.914]

Reading. On this targeted disagreement-focused slice, roberta-large-mnli aligns best with adjudicated labels by all three measures (Spearman $\rho = 0.446$, AUROC 0.794, AUPRC 0.929); RAGAS-style is second; the legacy DeBERTa proxy is weakly negative on correlation and below chance on AUROC. This reverses the typical-row ordering from the $n = 99$ calibration, where RAGAS-style ranked strongest. The combination supports the ControlledRAG claim that scorer ranking is evaluation-slice dependent, not a global ordering.

Caveat. The $n = 100$ batch is constructed to be adversarial for any single scorer (rows where the legacy DeBERTa proxy and the RAGAS-style judge disagree most). It is therefore not a population-level faithfulness estimate and not a substitute for the $n = 99$ random-sample calibration. The two slices answer two different audit questions, and ControlledRAG asks for both to be reported.

Source files. All under `human_eval_final/n100_disagreement/`:
`annotation_batch_disagreement_100.csv`, `human_eval_rater1.csv`,
`human_eval_rater2.csv`, `human_eval_adjudicated.csv` (rater 1, rater 2, adjudicated, notes),
`human_disagreement_summary.csv`, `human_disagreement_label_distribution.csv`,
`human_disagreement_agreement.csv`, `human_disagreement_correlations.csv`,
`human_disagreement_auroc_auprc.csv`, `human_disagreement_bootstrap_cis.csv`.

N Source Trace and Artifact Manifest

Every numerical claim in the main paper traces to a result file under `results/revision/` with a corresponding pre-specified script under `experiments/`. Table 23 summarises the trace; the full mapping is shipped as `SOURCE_TRACE.md` in the artifact and reproduced here for the reviewer’s convenience.

Table 23: Audit trace from claim to source file.

Audit	Source results files
Matched-context (§C backbone)	results/revision/fix_01/
Scaled audit (§B)	results/revision/fix_02/
Metric fragility (Mistral)	results/revision/fix_03/, standardized_scorer_fragility.csv
Threshold transfer (§K)	results/revision/fix_04/
Noise (§J)	results/revision/fix_05/
Cost-aware head-to-head	results/revision/fix_06/, h2h_summary_with_ci.csv, fix_11/
Span/control (§C)	results/revision/fix_12/
Stronger retrievers (§D)	results/revision/fix_13/, results/multi_retriever/
Second generator (§E)	results/revision/fix_14/
Long-form stress (§H)	results/revision/fix_15/
Human calibration $n = 99$ (§L)	human_eval_final/n99_calibration/
Targeted disagreement evaluation $n = 100$ (§M)	human_eval_final/n100_disagreement/

Reproducibility. An anonymized artifact package containing code, per-query CSVs, matched context pairs, source tables, scripts, and documentation is provided for review. Public package, benchmark, and permanent archive links are omitted during double-blind review and will be released upon acceptance.

O Compute Notes

The local scripts are resumable through their CSV outputs. The only script likely to benefit from a free-tier GPU notebook is the second-generator replication if local Ollama is slow: run the same command on a free GPU-backed notebook after pulling Qwen2.5 and Mistral into Ollama, then download data/revision/fix_14/ and results/revision/fix_14/. None of the audit experiments require paid APIs; all scorer rescoring is local DeBERTa, roberta-large-mnli, and a Mistral-via-Ollama RAGAS-style judge.

P Positioning Against Recent RAG Evaluation Work

The main paper compresses the related-work positioning into a single paragraph in the “Related Work, Continued” section. The detailed mapping is reproduced here for reviewer convenience.

Table 24: Positioning against recent RAG evaluation work.

Work	Primary contribution	ControlledRAG relation
ARES [Saad-Falcon et al., 2024]	trained judges for context relevance, answer faithfulness, and answer relevance with PPI calibration	complementary evaluator; ControlledRAG asks whether claims survive scorer/retriever/generator choices
RAGChecker [Ru et al., 2024]	fine-grained retrieval and generation diagnostics for RAG systems	complementary metric framework; ControlledRAG is the disclosure layer around such diagnostics
RAGTruth [Niu et al., 2024]	manually annotated hallucination corpus with word-level labels	complementary benchmark; ControlledRAG asks how benchmark conclusions transport across audit axes
MIRAGE [Park et al., 2025]	RAG benchmark with component-specific metrics and adaptability probes	complementary benchmark; ControlledRAG emphasizes reporting hidden degrees of freedom
FaithJudge [Tamber et al., 2025]	LLM-as-judge faithfulness benchmark and evolving leaderboard	complementary scorer/leaderboard; ControlledRAG requires scorer calibration and cross-scorer disclosure
RAGAS / TRUE / SummaC [Es et al., 2024, Honovich et al., 2022, Laban et al., 2022]	reference-free or NLI-style factual-consistency metrics	metrics used as audit inputs, not treated as reference standards

NeurIPS Paper Checklist

1. Claims

Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The paper title and contributions frame the work as a pre-registered audit and minimum reporting standard for RAG faithfulness evaluation. The abstract and introduction explicitly disclose: the matched-context CCS null; the small and seed-sensitive scaled headline (no-refinement / aggressive-refinement / gated-refinement, prior names HCPC-v1 / HCPC-v2); the fixed-generation scorer fragility and context-conditioned NLI sign-flip; the slice-dependent scorer—human ranking between the typical-row calibration ($n=99$) and the targeted disagreement slice ($n=100$); the threshold-transfer instabilities; and the cost-axis dominance of the head-to-head. The contribution is the standard, not a new method or leaderboard.

2. Limitations

Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: §12 is grouped into three categories: *audited surface* (mostly short-answer SQuAD, SQuAD-only matched-context, supplement-only 40-question long-form probe, mismatched Self-RAG harness); *human evaluation* (typical-row calibration $n=99$ with overlapping bootstrap CIs, targeted disagreement slice $n=100$ adversarial-by-construction, two raters plus adjudication on each); *scope of replication* (one local Qwen2.5 second-generator probe at $n=100$, context-conditioned NLI rescoring on the $n=600$ balanced subset not the full 7,500). The section ends with the explicit statement that these limits do not threaten the methodological claim because the seven-axis decomposition is motivated by the disclosure principle in §4.

3. Theory Assumptions and Proofs

For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper makes no formal theoretical claims requiring proofs. All claims are empirical or definitional.

4. Experimental Result Reproducibility

Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: §3 specifies the pipeline, embedder, indexer, reranker, generator, decoding settings, sample sizes, seed list, and statistical protocol. The supplementary material provides pre-specification, per-seed scaled-detail, and scorer-implementation details. The artifact ships per-query CSVs and scripts. *Reproducibility scope:* `run_all_analysis.sh` is end-to-end on a laptop in under two minutes for the verification path (it regenerates every analysis table in the main paper from the frozen per-query CSVs in `results/revision/` and `human_eval_final/`). Re-running the original *generation* and *scoring* pipelines (matched-context generation, scaled audit, second-generator probe, context-conditioned NLI rescoring) is a separate path that needs local Ollama plus a CPU/MPS or free-tier GPU notebook and takes hours; the supplementary compute notes document that path. The two-minute verification path covers all numerical claims in the main paper.

5. Open access to data and code

Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The reviewer-facing anonymized artifact (supplementary ZIP) contains all per-query CSVs, the human-evaluation final files, analysis scripts, the pre-specification log, the source trace, and a reproducibility README with verification commands. Public release links and DOIs will be added at acceptance.

715 **6. Experimental Setting/Details**

716 Does the paper specify all the training and test details (e.g., data splits, hyperparameters,
717 how they were chosen, type of optimizer, etc.)?

718 **Answer:** [Yes]

719 **Justification:** The audit holds the entire RAG pipeline fixed and varies only the audit axis
720 under test. Embedder, indexer, reranker, refinement policy, generator, and decoding settings
721 are listed in §3; sample sizes and seeds are listed for every audited cell; the matched-context
722 construction (similarity gap ≤ 0.02 , CCS gap ≥ 0.20) is given.

723 **7. Experiment Statistical Significance**

724 Does the paper report error bars suitably and correctly defined or other appropriate informa-
725 tion about the statistical significance of the experiments?

726 **Answer:** [Yes]

727 **Justification:** Every reported contrast is paired with a 10,000-resample bootstrap CI, paired
728 Wilcoxon p -value where applicable, Cohen’s d_z for paired contrasts, Wilson 95% CIs for
729 binary rates, and bootstrap CIs for the cross-scorer correlations. Multiple-comparison han-
730 dling: only the matched-context test is treated as the pre-specified primary; other paired
731 contrasts are descriptive across audit cells, as disclosed in §3.

732 **8. Experiments Compute Resources**

733 For each experiment, does the paper provide sufficient information on the computer resources
734 (type of compute workers, memory, time of execution) needed to reproduce the experiments?

735 **Answer:** [Yes]

736 **Justification:** The supplementary compute notes specifies the local Ollama / Apple-silicon
737 MPS / free-tier Kaggle T4 hardware, the per-fix runtime envelopes, and the analysis-only
738 verification path (run_all_analysis.sh, <2 min on any laptop). No paid APIs are
739 required.

740 **9. Code Of Ethics**

741 Does the research conform, in every respect, with the NeurIPS Code of Ethics?

742 **Answer:** [Yes]

743 **Justification:** The work uses only public datasets, free or local compute, no human-subjects
744 data beyond two short-answer support-judgment annotation passes by adult raters on public-
745 dataset content. No personally identifiable information, sensitive attributes, or facial /
746 biometric content is involved. No deceptive practices. Annotator identities are not stored in
747 the artifact.

748 **10. Broader Impacts**

749 Does the paper discuss both potential positive societal impacts and negative societal impacts
750 of the work performed?

751 **Answer:** [Yes]

752 **Justification:** ControlledRAG’s intended impact is to make RAG faithfulness claims more
753 honestly auditable, which reduces the risk that deployed RAG systems are fielded based on
754 under-identified single-cell numbers (negative impact: false confidence in deployed medical
755 / legal RAG; the standard is meant to mitigate that). The work is descriptive; it does not
756 introduce a new generative model that could itself be misused.

757 **11. Safeguards**

758 Does the paper describe safeguards that have been put in place for responsible release of
759 data or models with a high risk for misuse?

760 **Answer:** [N/A]

761 **Justification:** The paper does not release a new generative model, dataset of personally
762 identifiable content, or other artifact with high misuse risk. The released artifacts are
763 evaluation tables, scripts, and short-answer adjudicated annotation labels on public data;
764 these do not present such risks.

765 **12. Licenses for existing assets**

766 Are the creators or original owners of assets (e.g., code, data, models), used in the paper,
767 properly credited and are the license and terms of use explicitly mentioned and properly
768 respected?

769 **Answer:** [Yes]

770 **Justification:** The paper cites SQuAD, PubMedQA, HotpotQA, NaturalQuestions, Triv-

771 iaQA, QASPER, MS-MARCO, and the open embedder/cross-encoder/NLI models used.
772 Each is used under its published research-use license. No proprietary asset is used.

773 **13. New Assets**

774 Are new assets introduced in the paper well documented and is the documentation provided
775 alongside the assets?

776 **Answer:** [Yes]

777 **Justification:** The new released assets are (i) the ControlledRAG seven-
778 axis reporting standard (§4) plus a filled-in reporting-checklist template
779 (controlledrag_reporting_checklist.md, distinct from this mandatory NeurIPS
780 checklist), (ii) per-query CSVs for every audited cell, (iii) the typical-row calibra-
781 tion set ($n=99$), (iv) the targeted disagreement slice ($n=100$, two raters plus ad-
782 judication, 74/26/0 adjudicated distribution), and (v) the analysis scripts. Docu-
783 mentation: README_REPRODUCE.md, SOURCE_TRACE.md, artifact_manifest.md,
784 CLAIMS_AUDIT.md, and the supplementary source-trace section.

785 **14. Crowdsourcing and Research with Human Subjects**

786 For crowdsourcing experiments and research with human subjects, does the paper include
787 the full text of instructions given to participants and screenshots, if applicable, as well as
788 details about compensation (if any)?

789 **Answer:** [Yes]

790 **Justification:** The two adjudicated human-annotation arms ($n=99$ typical-row calibration;
791 $n=100$ targeted scorer-disagreement evaluation) are documented with their exact label
792 scheme, rater instructions, decision codebook, and adjudication template; these documents
793 are shipped in the artifact under human_eval_final/. No crowdsourcing platform was
794 used; no compensation was paid; the raters were not subject to risk.

795 **15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human**
796 **Subjects**

797 Does the paper describe potential risks incurred by study participants, whether such risks
798 were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals
799 (or an equivalent approval/review based on the requirements of your country or institution)
800 were obtained?

801 **Answer:** [N/A]

802 **Justification:** The annotation tasks are short support/faithfulness judgments on public-
803 dataset content (no personally identifiable information, no sensitive attributes, no risk-of-
804 harm content). Per common interpretations of the relevant human-subjects regulations, this
805 evaluation does not constitute research with human subjects requiring IRB review. The
806 annotators were informed of the purpose of the labelling.

807 **16. Declaration of LLM usage**

808 Does the paper describe the usage of LLMs if it is an important, original, or non-standard
809 component of the core methods in this research? Note that if the LLM is used only for
810 writing, editing, or formatting purposes and does not impact the core methodology, scientific
811 rigoroussness, or originality of the research, declaration is not required.

812 **Answer:** [Yes]

813 **Justification:** A local Mistral-7B (via Ollama) is used as both the audited generator and
814 as the RAGAS-style local judge in the metric-fragility audit. A local Qwen2.5-7B is used
815 as the second-generator replication. Their use, decoding settings, and the fact that the
816 RAGAS-style judge is itself an audit input (not a ground-truth oracle) are disclosed in §3
817 and §7. LLMs are research artifacts in the audited pipeline, not ground-truth oracles or
818 co-authors.